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May 6, 2026

The Editor-in-Chief
Journal Name
Publisher

Dear Editor,

We are pleased to submit our manuscript entitled **"Bilevel Optimization for Supply Chain Network Design under Carbon Taxation"** for consideration for publication in *Journal Name*.

Scope and novelty. This paper addresses the strategic interaction between a government setting a carbon tax rate and a supply chain operator designing a closed-loop network in response. We propose a bilevel mixed-integer programming model that explicitly captures this Stackelberg leader–follower structure. To our knowledge, this is the first work to combine a bilevel carbon taxation framework with a closed-loop supply chain network design problem including integer facility decisions.

Methodological contribution. We develop a Deterministic Bounding Procedure (DBP) with proven convergence guarantees and provide reproducible computational artifacts for the implemented DBP workflows.

We confirm that:

- This manuscript has not been published elsewhere and is not under consideration by another journal.
- All authors have approved the final version and agree with the submission.
- We have no conflicts of interest to declare.

We suggest the following potential reviewers with expertise in bilevel optimisation and sustainable supply chains:

- Reviewer A (University X): reviewer.a@univ.edu
- Reviewer B (University Y): reviewer.b@univ.edu
- Reviewer C (University Z): reviewer.c@univ.edu

Thank you for considering our submission. We look forward to your response.

Yours sincerely,

Author One
(on behalf of all co-authors)